

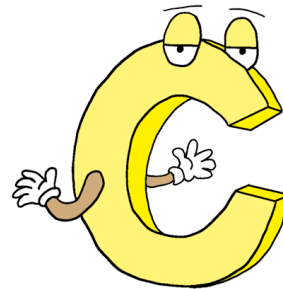
C Programming

FYD401

```
17  
18 void loop()  
19 {  
20     //MCU Task  
21     for(NUM_FN_TASK_CNT = 0; ((NUM_FN_TAS  
22     {  
23         if ((millis() - fn[NUM_FN_TASK_O  
24         {  
25             fn[NUM_FN_TASK_CNT].time_cnt =
```



Oh! I



Credits

- The slides for the course were prepared by Andreas Heinz (Chalmers University of Technology)
 - Minor modifications were made
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General Information (I)

- Teacher
 - Paolo Vinai
 - vinai@chalmers.se
 - Teaching assistants
 - Hampus Hansen
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 - Fredrik Öhrlund
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General Information (II)

FDY401 course in CANVAS

<https://canvas.gu.se/courses/100255>

If you are registered, login information for the CANVAS room of the course is available in LADOK

Course Schedule

- 6 lectures
 - 2 hours each, h. 18:00-20:00
 - 9 exercise sessions = examination
 - 4 hours each, h. 18:00-22:00
 - Link to the schedule
 - <https://cloud.timeedit.net/chalmers/web/public/ri177150X55Z5QQ6Z96g3510y00Y6Y75505gQY6Q539.html>
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Lectures

Lecture



Exam



- Lectures
 - They provide only a brief introduction to programming
 - Support for learning and examples
- NO substitute for practical work and reading!

Lecture slides will be posted on Canvas after each lecture (see section 'Files').

In theory there is no difference between theory and practice; in practice there is.

Yogi Berra

Everything is practice.

Pele

Course Literature

- Book: "Vägen till C" by Bilting & Skansholm
 - Many other books ...
 - Many online resources ...
 - Some suggestions in Canvas ...
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Examination (I)

- Exercises and instructions in the compendium
 - <https://canvas.gu.se/courses/100255/files/folder/Exercises>
 - Students present the solution of the exercises in the exercise sessions
 - Alone or in teams of maximum 2 students
 - Procedure followed in the exercise sessions
 - <https://canvas.gu.se/courses/100255/pages/procedure-for-the-exercise-sessions>
 - Requirements to have exercises approved
 - working programs, good programming practice, understanding of the material, proper I/O, comments
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Examination (II)

- End of the course = end of the examination
 - Wednesday, October 28, 2026
 - Students get points for each exercise approved
 - At least one exercise per sub-session
 - Grading
 - G = between 30 and 39 points in total
 - VG = at least 40 points in total
 - <https://canvas.gu.se/courses/100255/pages/requirements-to-pass-the-course>
-

Software

- Visual Studio Code
 - <https://code.visualstudio.com/>
 - There are alternatives ...
 - If you prefer a different code editor or integrated development environment, it will still be fine
 - <https://canvas.gu.se/courses/100255/pages/software-for-the-course>
-

Preliminary Lecture Plan

Introduction: General course information

Lecture 1: Computers and programming, history and development of C, a simple program
[Vägen till C, Chapter 1 & 2]

Lecture 2: Vocabulary, Algorithms, Types, Compilation

Lecture 3: Preprocessor statements, standard functions, expressions, operators [Chapter 4]

Lecture 4: Control structures and statements, functions and modular programming, pointers
[Ch. 6 & 7]

Lecture 5: Pointers, arrays, dynamic memory allocation, structs [Chapters 7 & 8]

Lecture 6: Structured variables, string handling, files and streams, linking, errors [Chapter 10]

Teach yourself variables, constants, arrays, strings, expressions, statements, functions....



Teach yourself program flow, pointers, references, classes, objects, inheritance, polymorphism,



Do a lot of recreational programming. Have fun hacking but remember to learn from your mistakes.



Interact with other programmers. Work on programming projects together. Learn from them.



Teach yourself advanced theoretical physics and formulate a consistent theory of quantum gravity.



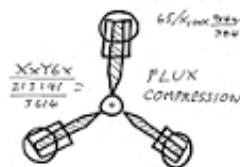
Teach yourself biochemistry,
molecular biology, genetics,...



Use knowledge of biology to make an age-reversing potion.



Use knowledge of physics to build flux capacitor and go back in time to day 21.



Replace younger self.



Works similar for C

As far as I know, this is the easiest way to "Teach Yourself C++ in 21 Days".

Do you have questions about the course?
